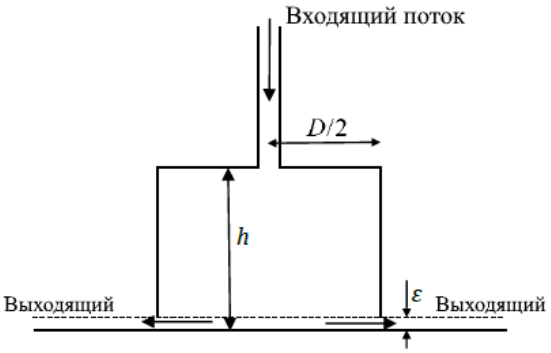
Air cushion (10 points)

Air cushions are often used to move heavy loads. In this problem we study how one works; the principle is shown schematically below. Under the platform there is a cylindrical space of diameter $D = 40$ cm and height $h = 10$ cm into which air is pumped. The ambient temperature is $T_0 = 300$ K and the atmospheric pressure $P_{atm} = 1.01 \cdot 10^5$ Pa. The weight of the platform and its load is $W = 78.48$ kN. When the pressure force of the air exceeds it, the platform lifts and leaves a gap of width ε above the ground. The compressor keeps a constant air flow.

Air may be treated as an ideal gas; the flow is adiabatic and quasi-static. We are mainly interested in steady states. For numbers use the molar mass of air $\mu = 29.0$ g/mol and $\gamma = c_P/c_V = 1.4$. For the specific heats $c_P - c_V = r$, where $r = 287\ \mathrm{J/(kg \cdot K)}$, and the equation of state reads $P = \rho r T$.



Part A. Equilibrium of the platform (1 point)

A.1	Express c_V and c_P in terms of r and γ. Find their values.	0.5pt

A.2	How does the ratio of the pressure in the cylindrical cavity to the atmospheric pressure, P_0/P_{atm}, depend on the weight of the platform and the dimensions of the system?	0.5pt

Part B. How an air cushion works (5.5 points)

The mass flow rate of the gas escaping through the gap at the ground is $\mathrm{d}m/\mathrm{d}t = \rho v A$, where v and ρ are the velocity and density of the escaping gas.

Consider a jump of density and pressure propagating with speed c in a tube of unit cross-section. The disturbed gas moves with speed v, the undisturbed gas is at rest. The pressure and density are P, ρ in the disturbed region and P_0, ρ_0 in the undisturbed one, whose temperature is T. Some of the reasoning is easier in the frame in which the wave front (the boundary of the undisturbed region) is at rest.

B.1	Express the velocity v of the disturbed gas in terms of c, ρ and ρ_0.	0.5pt

B.2	Express the pressure difference $P - P_0$ in terms of ρ, ρ_0 and c.	1.0pt

B.3	Express c in terms of γ, r, T.	0.5pt
B.4	Calculate the speed of sound in air at the ambient temperature T_0.	0.5pt
B.5	When the gas escapes from under the platform it expands adiabatically and its temperature T differs from T_0. Introduce the Mach number $M = v/c$, where c is the speed of sound in the escaping gas, so that $\mathrm{d}m/\mathrm{d}t = \rho A M c$. Express $\mathrm{d}m/\mathrm{d}t$ in terms of M, γ, T/T_0 and $P_0 A\gamma/c_0$, where c_0 is the speed of sound at T_0.	1.0pt
B.6	Express the gas velocity v in terms of M, T, r and γ.	0.5pt
B.7	Using Bernoulli's equation for a compressible gas, $\frac{v^2}{2} + c_P T = \text{const}$, find the ratio T/T_0 in terms of M and γ.	1.5pt

Part C. Critical flow (2.0 points)

It can be shown that when gas flows from a narrow passage into a wide one, its velocity cannot exceed the local speed of sound ($M \leq 1$). As the pumped flow changes, so does the exit velocity v; for large enough flows it may reach its maximum value c.

C.1	Express $\mathrm{d}m/\mathrm{d}t$ in terms of M, γ and $P_0 A\gamma/c_0$ and sketch it as a function of $M \in [0;1]$. Find the value of M at which the flow is maximal (the critical regime).	1.0pt
C.2	Let the compressor deliver $5.0\ \mathrm{m}^3/\mathrm{min}$ (of atmospheric air). Calculate ε, T, P, v and ρ of the escaping flow in the critical regime.	1.0pt

Part D. Uneven ground (1.5 points)

In practice the roughness of the ground does not let the gap ε exceed 0.5 mm, so the critical regime cannot be reached.

D.1	Calculate M, T, P, v and ρ of the escaping gas for $\varepsilon = 0.5$ mm and $\mathrm{d}m/\mathrm{d}t = 0.1$ kg/s.	1.5pt